

Deceptive Previews: A Study of the Link Preview Trustworthiness in Social Platforms

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Social media has become a primary mean of content and information sharing, thanks to its speed and simplicity. In this scenario, link previews play the important role of giving a meaningful first glance to users, summarizing the content of the shared webpage within its title, description and image. In our work, we analyzed the preview-rendering process, observing how it is possible to misuse it to obtain benign-looking previews for malicious links. Concrete use-case of this research field is phishing and spam spread, considering targeted attacks in addition to large-scale campaigns.

We designed a set of experiments for 20 social media platforms including social networks and instant messenger applications and found out how most of the platforms follow their own preview design and format, sometimes providing partial information. Four of these platforms allow preview crafting so as to hide the malicious target even to a tech-savvy user, and we found that it is possible to create misleading previews for the remaining 16 platforms when an attacker can register their own domain. We also observe how 18 social media platforms do not employ active nor passive countermeasures against the spread of known malicious links or software, and that existing cross-checks on malicious URLs can be bypassed through client- and server-side redirections. To conclude, we suggest seven recommendations covering the spectrum of our

findings, to improve the overall preview-rendering mechanism and increase users' overall trust in social media platforms.

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